

HEALTHCONNECT FINAL ANALYTICS & DECISION SUPPORT PACKAGE

Data Analytics Track

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Project	Improving Patient Appointment Attendance and Healthcare Support Using Data and AI
Track	Data Analytics

1. Executive Summary

HealthConnect Clinic faces a 48.5% no-show rate across 5,000 appointments, indicating that a substantial share of booked appointment capacity is not being used. The Week 6 dashboard was independently validated against the raw CSV using Python (Jupyter Notebook), with all reported validation checks passing and zero variance against the validated KPI calculations.

The strongest analytical driver identified is distance from the clinic: the Very Far group recorded 68.1% no-shows compared with 46.5% for the Close group, a 21.6 percentage-point spread. Booking lead time further compounds this pattern, with the Very Far + 2–4 Weeks segment recording an 83.3% no-show rate across the four appointment types.

Reminder activity is associated with a 4.9 percentage-point attendance difference (47.6% with a reminder versus 42.7% without). SMS recorded 49.6% attendance overall, while Email narrowly exceeded SMS within the Moderate Distance band (47.1% versus 46.4%), providing a specific opportunity for controlled A/B testing.

Cross-track validation with Data Science further strengthened the solution. Dorsilla Kemunto's model independently identified Booking Lead Time and Distance as the top two predictive features. After removal of the **Unnamed: 0** artefact column, the top features remained unchanged. Both tracks now use the same cleaned 5,000-record CSV, improving consistency between dashboard analysis and model development.

2. Final KPIs — Confirmed

KPI	Value	Validation
No-Show Rate	48.5%	PASS
Attendance Rate	46.3%	PASS
Attendance with Reminder	47.6%	PASS

Attendance without Reminder	42.7%	PASS
Reminder Effectiveness Gap	4.9 pp	PASS
Very Far Distance No-Show	68.1%	PASS
Close Distance No-Show	46.5%	PASS
Highest-Risk Combo (Very Far + 2–4 Weeks)	83.3%	PASS
SMS Attendance	49.6%	PASS

3. Validated Key Findings

1. Overall No-Show Rate: 48.5% — independently recalculated and verified against the raw CSV.
2. Reminder Association: attendance was 47.6% when a reminder was recorded versus 42.7% without a reminder, a 4.9 percentage-point difference.
3. Distance: Very Far patients recorded 68.1% no-shows compared with 46.5% for Close patients, creating a 21.6 percentage-point spread.
4. Combined Risk: Very Far + 2–4 Weeks produced an 83.3% no-show rate and the pattern held across Diagnostic, Follow-up, General, and Specialist appointment types.
5. Reminder Channel: SMS recorded 49.6% attendance overall. Within the Moderate Distance band, Email recorded 47.1% versus 46.4% for SMS; the small 0.7 percentage-point difference should be tested rather than treated as conclusive.
6. Age: age group alone showed a relatively narrow no-show range of 45.1%–50.8%, suggesting that age is more useful as an interaction variable with distance and booking lead time than as a standalone intervention lever.

4. Business Insights

- Reminder interventions are associated with a meaningful but modest improvement in attendance and should be retained as part of the attendance strategy.
- Distance from the clinic is the strongest single analytical driver identified in this dataset.
- Combining distance with booking lead time creates a more specific high-risk segment than either variable alone.
- SMS can serve as the default reminder channel based on overall attendance in this dataset, while Email should be tested specifically within the Moderate Distance segment.
- Age appears more informative when considered together with other risk factors rather than as a standalone segmentation variable.

5. Recommendations

Priority	Recommendation	Evidence / Rationale
High	Target the Very Far + 2–4 Weeks segment with intensified reminders and, where feasible, transport-support	83.3% no-show rate

	interventions.	
High	Use SMS as the default reminder channel and run a controlled A/B test of Email within the Moderate Distance band.	SMS: 49.6% overall attendance; Email: 47.1% vs SMS: 46.4% in Moderate Distance
Medium	Encourage shorter booking lead times where clinically and operationally feasible.	Short Term: 66.6% no-show vs 2–4 Weeks: 51.7% no-show
Medium	Evaluate telehealth or transport-support options for Very Far patients where clinically appropriate.	Distance is the strongest identified analytical driver
Low	Consider age-tailored messaging for the 18–24 and 55–64 groups, while monitoring interaction with distance and lead time.	These groups showed elevated no-show patterns

6. Analytical Limitations

- The Moderate Distance channel comparison is narrow: Email exceeds SMS by only 0.7 percentage points. The difference should therefore be treated as a test hypothesis rather than a definitive channel conclusion.
- The Not Specified distance group has a 52.2% no-show rate and may contain an unobserved distance-related pattern.
- Cross-page navigation and filter interaction have not yet been fully tested on the mobile layout.
- The Data Science model has not yet been rerun specifically on the refined Week 7 dashboard outputs.

7. Cross-Track Contribution

7.1 Shared with Data Science — Dorsilla Kemunto

- KPI baseline: No-Show Rate 48.5%; Attendance Rate 46.3%.
- Distance finding: Very Far Distance = 68.1% no-show.
- Reminder channel finding: SMS = 49.6% attendance.
- Highest-risk combination: Very Far + 2–4 Weeks = 83.3% no-show.

7.2 Received from Data Science

- Top two model features: Booking Lead Time and Distance.
- Confirmation that removing the Unnamed: 0 artefact column did not change the top two features.
- Model error distribution: True Negative 27.32%, True Positive 25.63%, False Negative 25.53%, False Positive 21.52%.

What changed: Both tracks now use a single cleaned CSV containing 5,000 records. The artefact column was removed, and the dashboard analysis and model inputs are now aligned.

Impact on HealthConnect: The model no longer relies on an artefact column, while the dashboard findings and model feature behaviour are based on the same cleaned dataset. This provides a stronger foundation for end-to-end analytical validation.

8. HC-POD Integration Matrix — Data Analytics → Data Science

#	Item	Detail
1	Track collaborated with	Data Science — Dorsilla Kemunto
2	Dependency	Model feature-importance validation against the analytical drivers identified in the dashboard
3	Output received	Top two model features: Booking Lead Time and Distance; model error distribution; confirmation that removing Unnamed: 0 did not change the top features
4	Output provided	Validated KPI baselines; Very Far distance finding (68.1% no-show); SMS finding (49.6% attendance); highest-risk combination (83.3% no-show)
5	Final integration activity	Compared the top analytical drivers with model feature importance; Data Science reran the model without Unnamed: 0; both tracks aligned on the same cleaned CSV
6	What changed	Artefact column removed; both tracks aligned on 5,000 records; dashboard and model inputs are consistent
7	How it improved the solution	Reduced the risk of model dependence on a non-analytical artefact and strengthened consistency between dashboard insights and model behaviour
8	Evidence	Week 7 validation table with PASS results; Data Science feature-importance output; error-distribution table; 5,000-record dataset alignment
9	Contribution to final presentation	Joint slide demonstrating that the Data Analytics top drivers and Data Science

		top model features both identify Booking Lead Time and Distance, with cross-track consistency confirmed
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9. Decision Support Summary

Priority action areas for HealthConnect: focus intervention on patients who live Very Far from the clinic and book 2–4 Weeks in advance; retain reminder use; maintain SMS as the overall default while testing Email in the Moderate Distance segment; and evaluate operational options such as shorter booking windows, telehealth, or transport support where appropriate.

Validation status: The Week 6 analytical outputs have been cross-checked against the raw CSV, the reported KPI validation checks passed, and the Data Analytics and Data Science tracks are aligned on the same cleaned 5,000-record dataset.